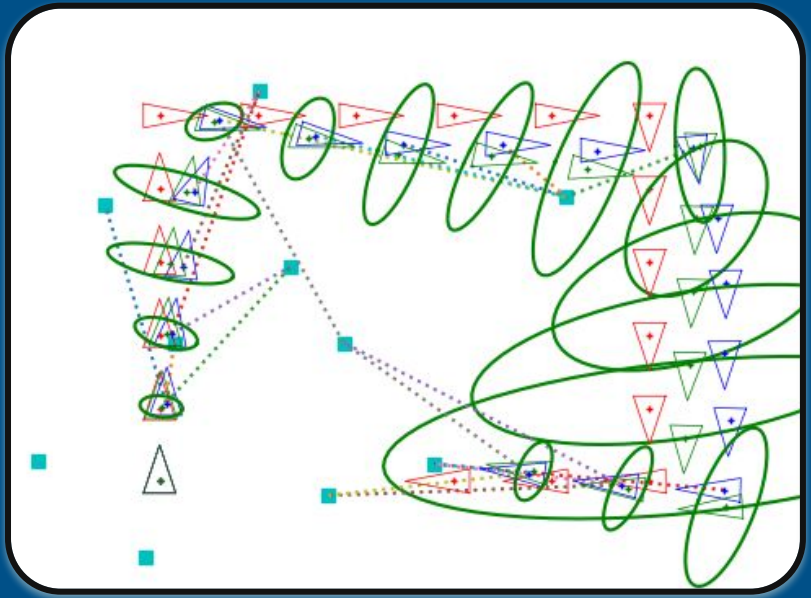
I402: Principios de la Robótica Autónoma

Clases Prácticas
Tutorial 6



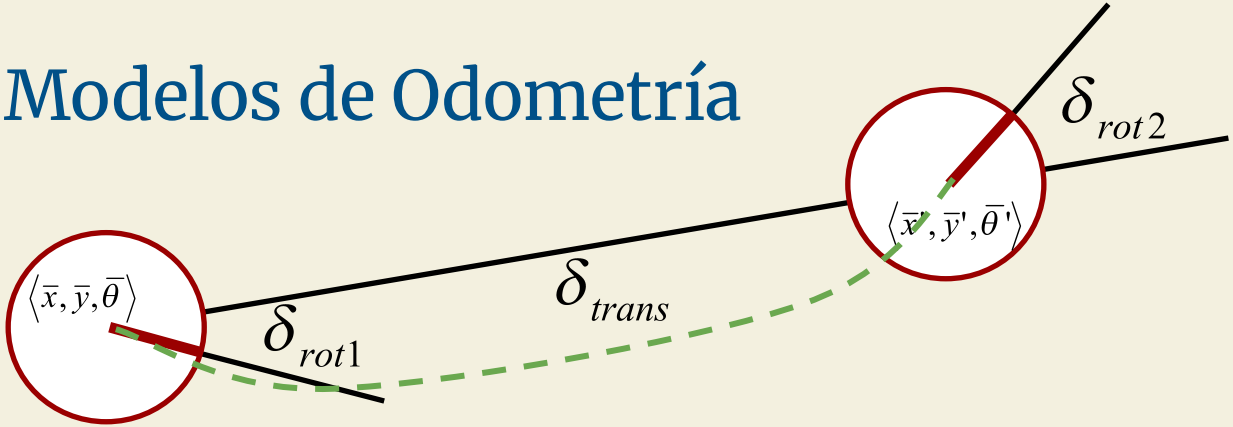
Universidad de SanAndrés

$$Bel(x_t) = \eta P(z_t \mid x_t) \int \boxed{P(x_t \mid u_t, x_{t-1})} Bel(x_{t-1}) dx_{t-1}$$

Modelos de Movimiento

$$P(x_t \mid u_t, x_{t-1})$$

Modelos de Odometría



Modelos de Odometría

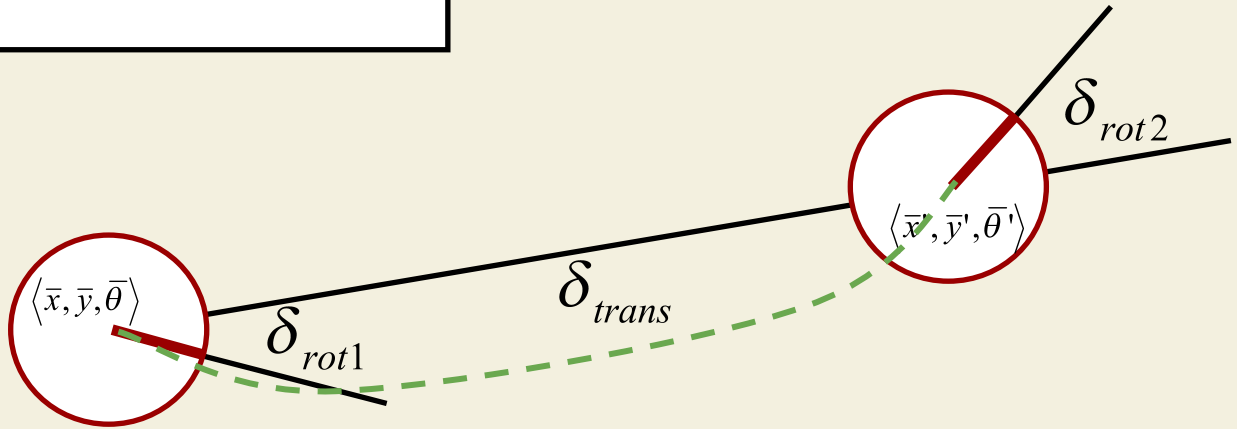
$$P(x_t \mid u_t, x_{t-1})$$

$$\delta_{trans} = \sqrt{(\bar{x}'-\bar{x})^2 + (\bar{y}'-\bar{y})^2}$$

$$\delta_{rot1} = \operatorname{atan2}(\bar{y}'-\bar{y}, \bar{x}'-\bar{x}) - \bar{\theta}$$

$$\delta_{rot2} = \bar{\theta}'-\bar{\theta} - \delta_{rot1}$$

La odometría me da, en base al movimiento de las ruedas, un Δx, Δy, $\Delta\theta$



¿Cómo me afecta el ruido?

Muestrear el modelo de Odometría

Algorithm sample_motion_model_odometry(u_t, x_{t-1})**:**

$$\delta_{\text{rot1}} = \text{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta}$$

$$\delta_{\text{trans}} = \sqrt{(\bar{x} - \bar{x}')^2 + (\bar{y} - \bar{y}')^2}$$

$$\delta_{\text{rot2}} = \bar{\theta}' - \bar{\theta} - \delta_{\text{rot1}}$$

$$\hat{\delta}_{\text{rot1}} = \delta_{\text{rot1}} - \textbf{sample}(\alpha_1 \delta^2_{\text{rot1}} + \alpha_2 \delta^2_{\text{trans}})$$

$$\hat{\delta}_{\text{trans}} = \delta_{\text{trans}} - \textbf{sample}(\alpha_3 \delta^2_{\text{trans}} + \alpha_4\ \delta^2_{\text{rot1}} + \alpha_4\ \delta^2_{\text{rot2}})$$

$$\hat{\delta}_{\text{rot2}} = \delta_{\text{rot2}} - \textbf{sample}(\alpha_1 \delta^2_{\text{rot2}} + \alpha_2 \delta^2_{\text{trans}})$$

$$x' = x + \hat{\delta}_{\text{trans}}\ \cos(\theta + \hat{\delta}_{\text{rot1}})$$

$$y' = y + \hat{\delta}_{\text{trans}}\ \sin(\theta + \hat{\delta}_{\text{rot1}})$$

$$\theta' = \theta + \hat{\delta}_{\text{rot1}} + \hat{\delta}_{\text{rot2}}$$

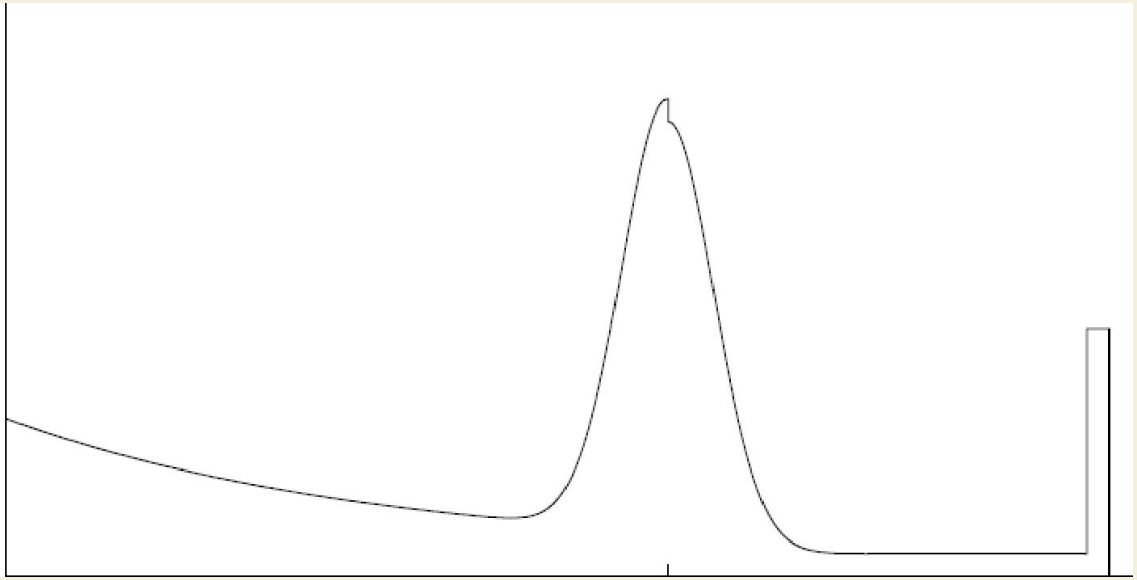
return $x_t = (x', y', \theta')^T$

OJO

Varianza

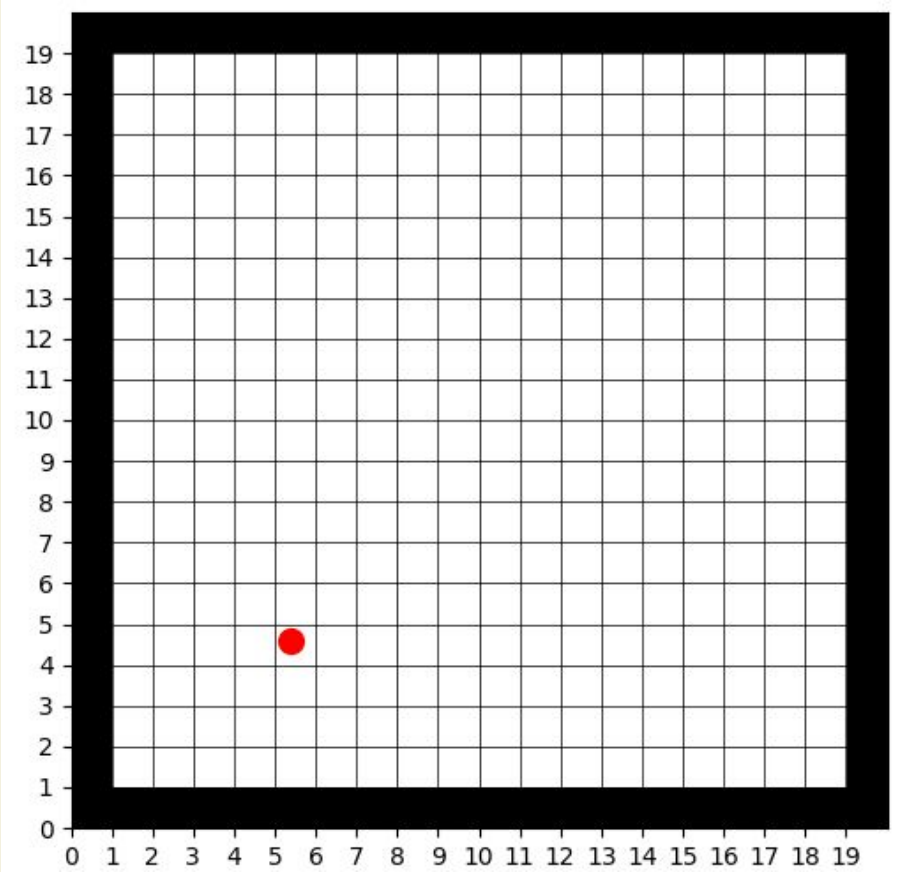
$$Bel(x_t) = \eta \boxed{P(z_t \mid x_t)} \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Modelos de Mediciones



¿Cómo calculo el punto de contacto?

Bresenham Lines



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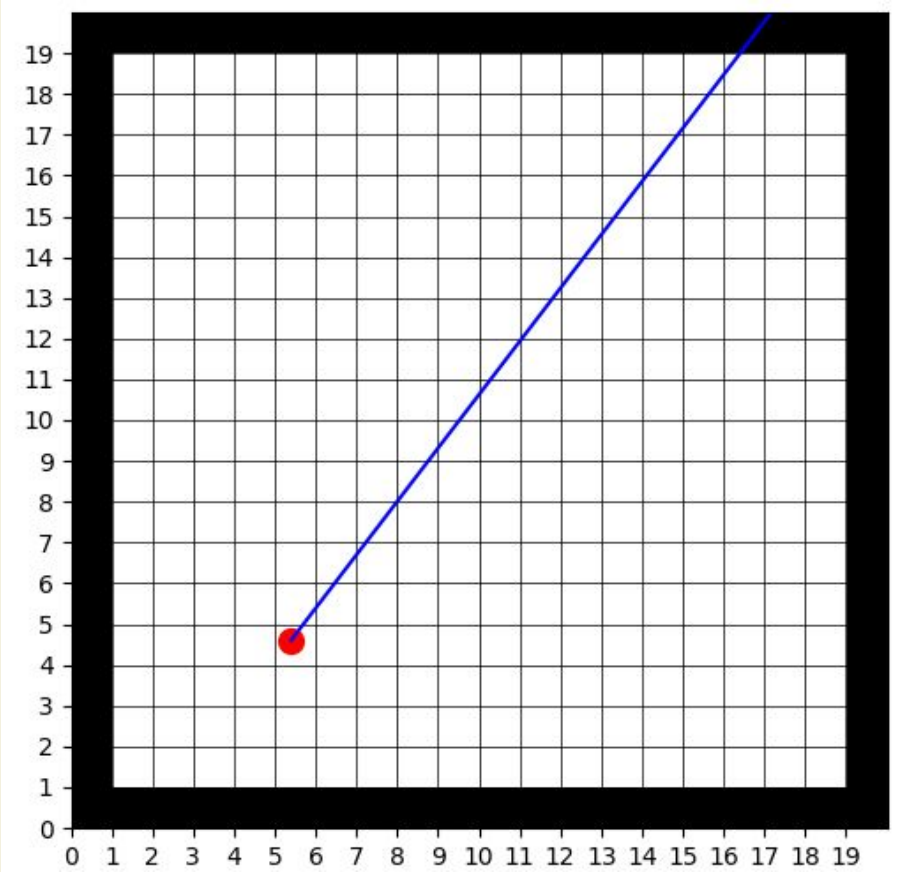

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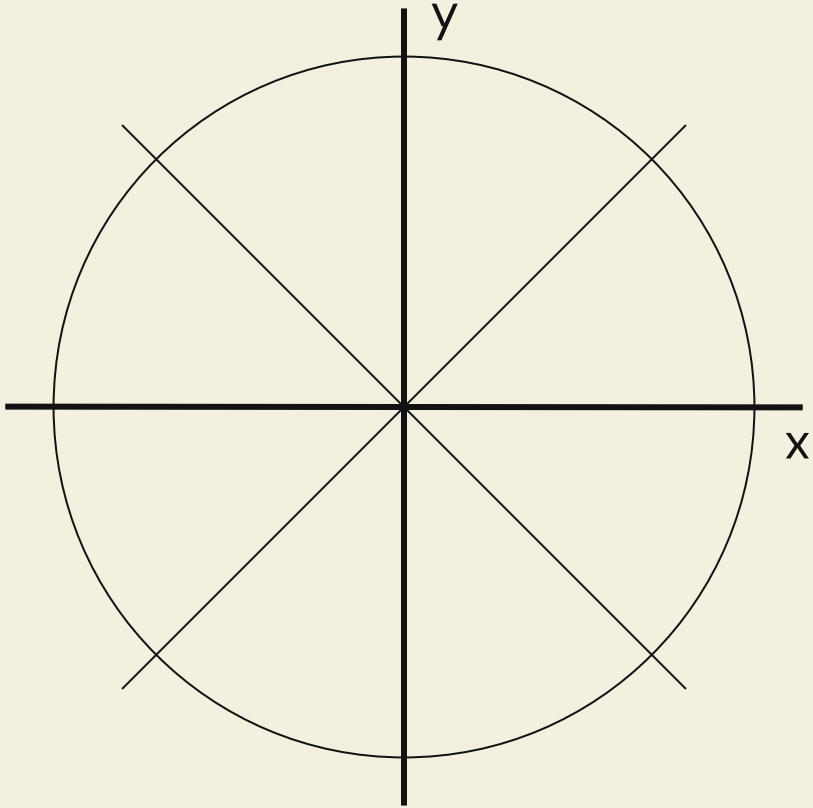

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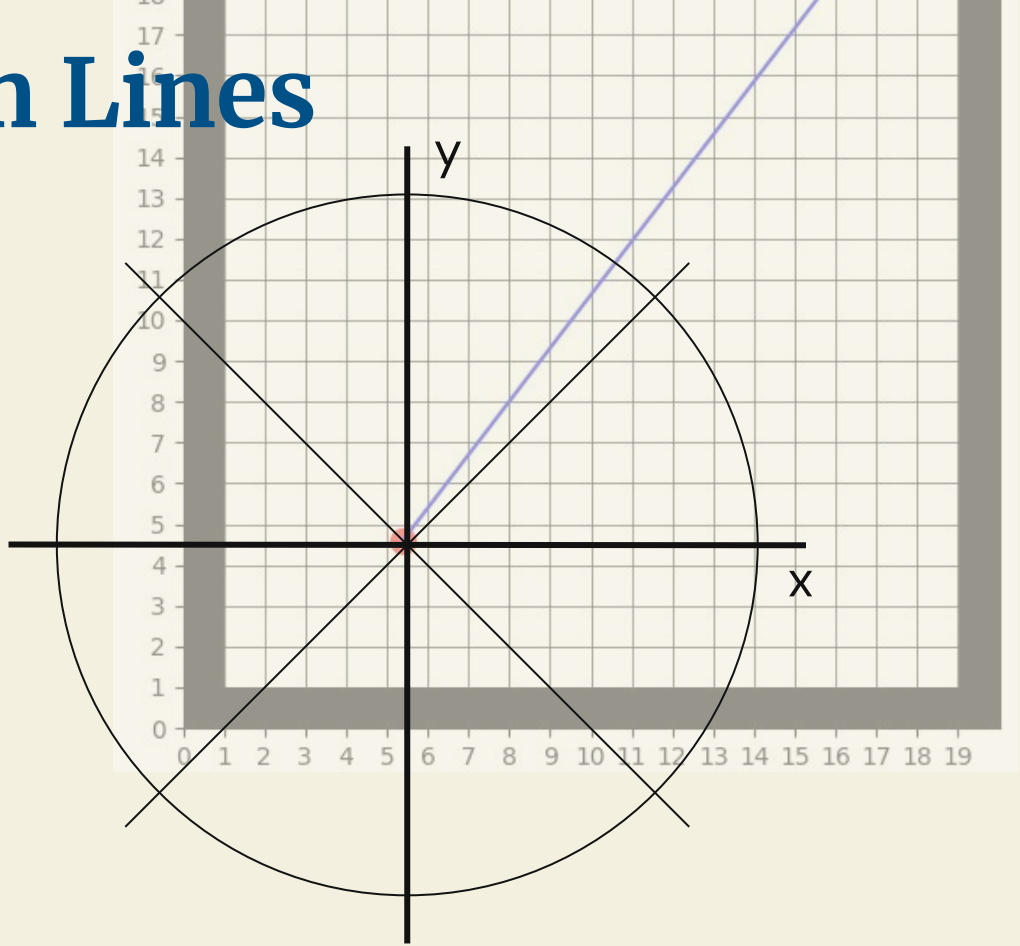

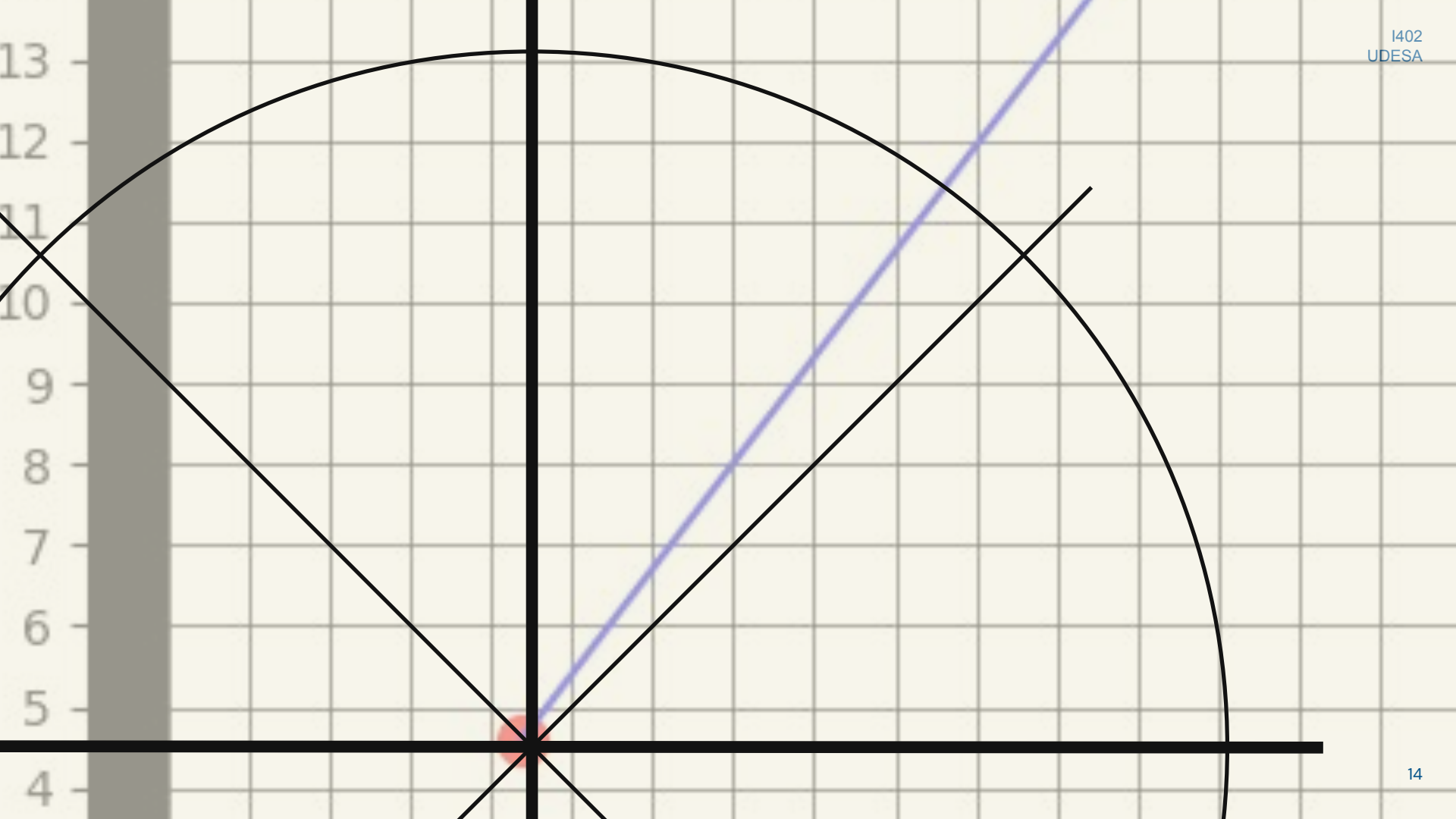
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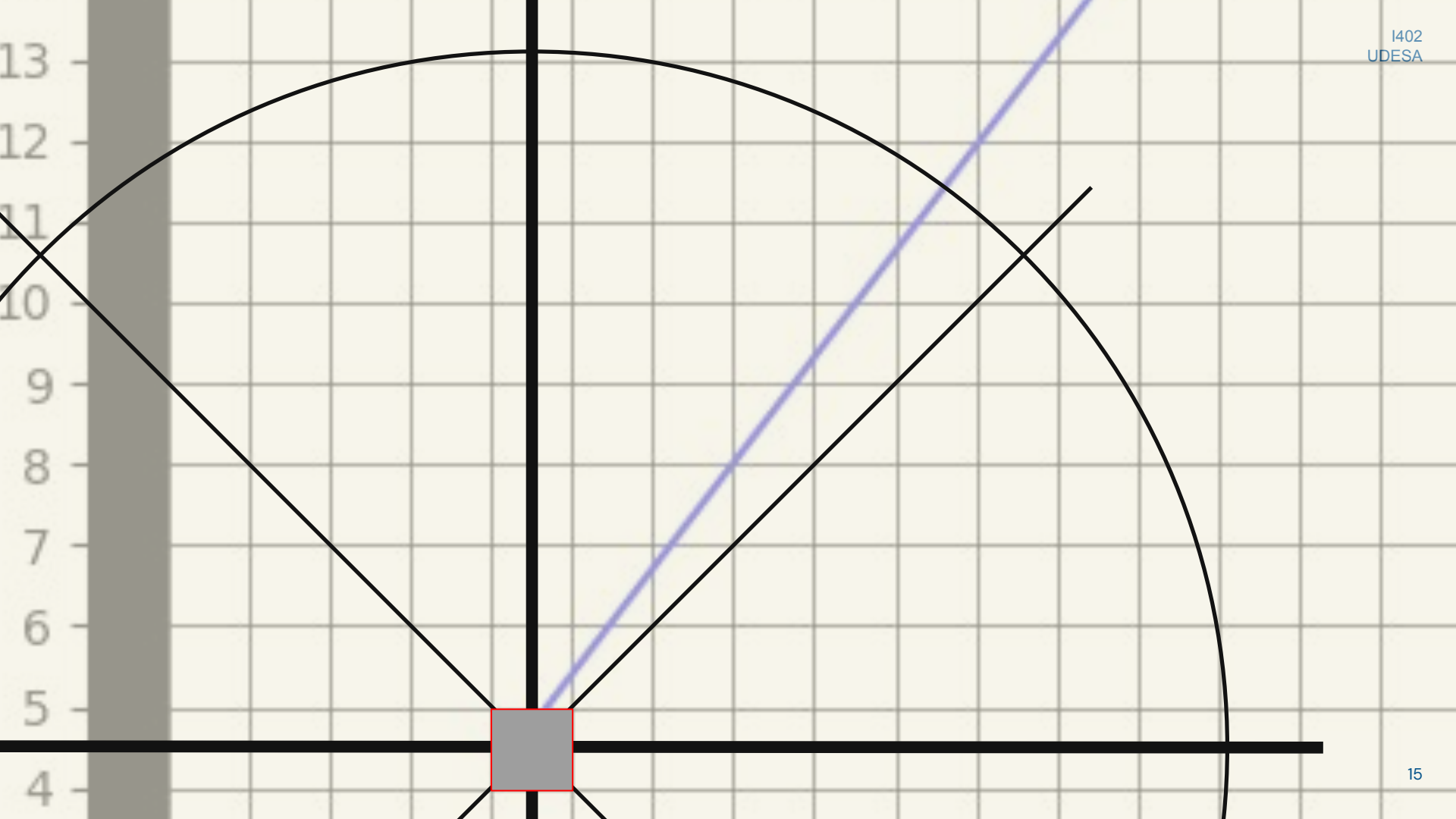
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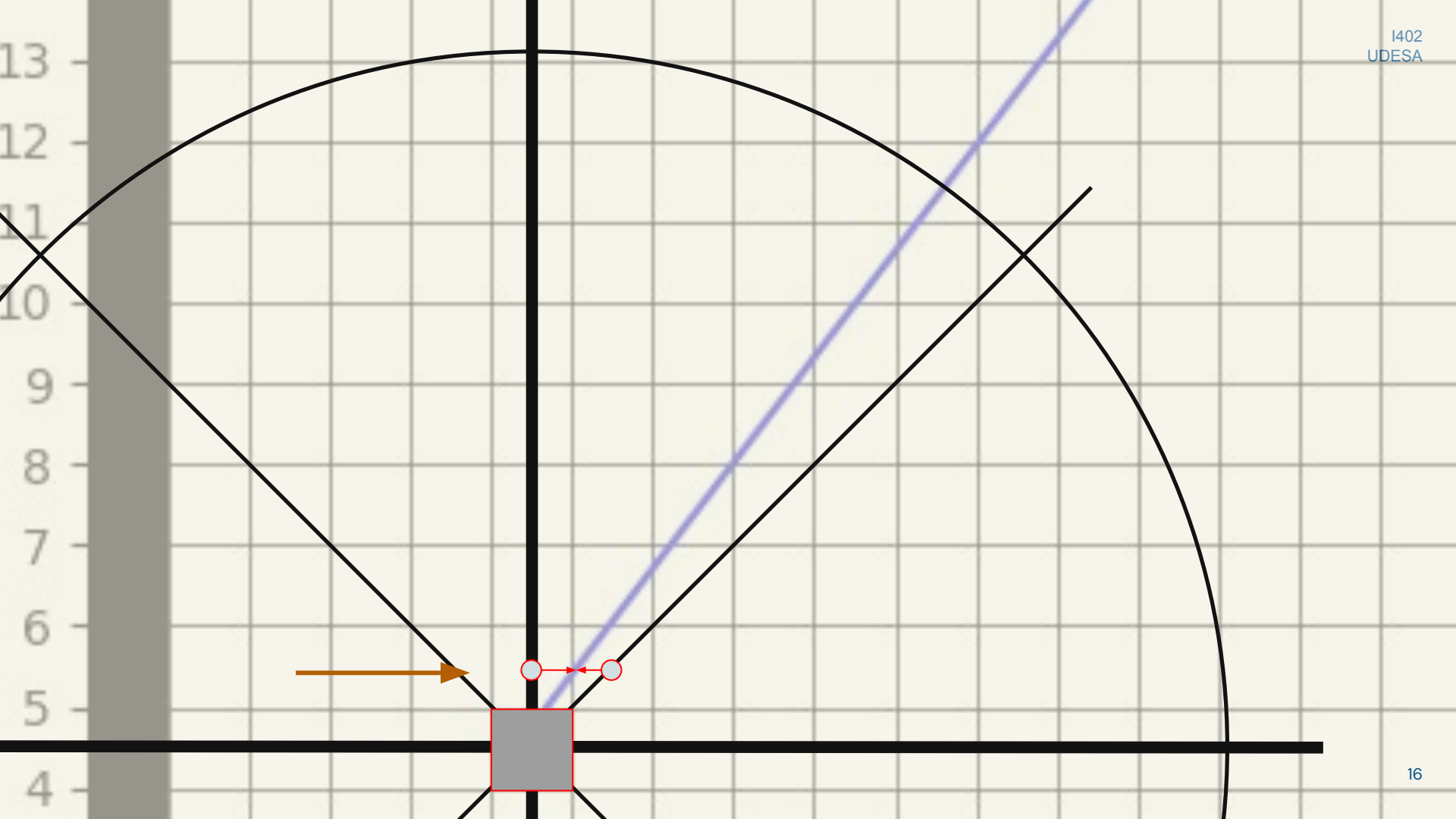
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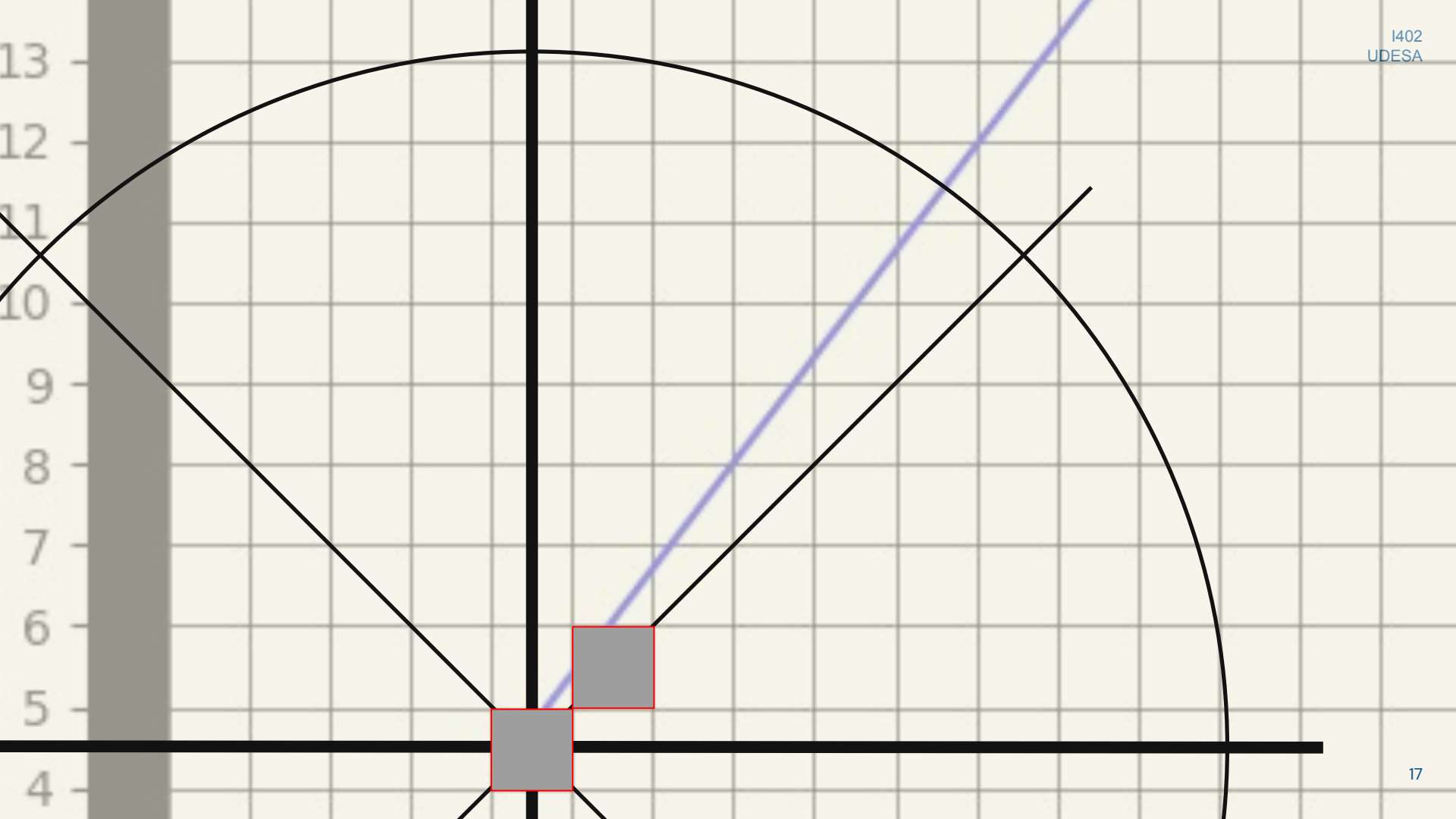
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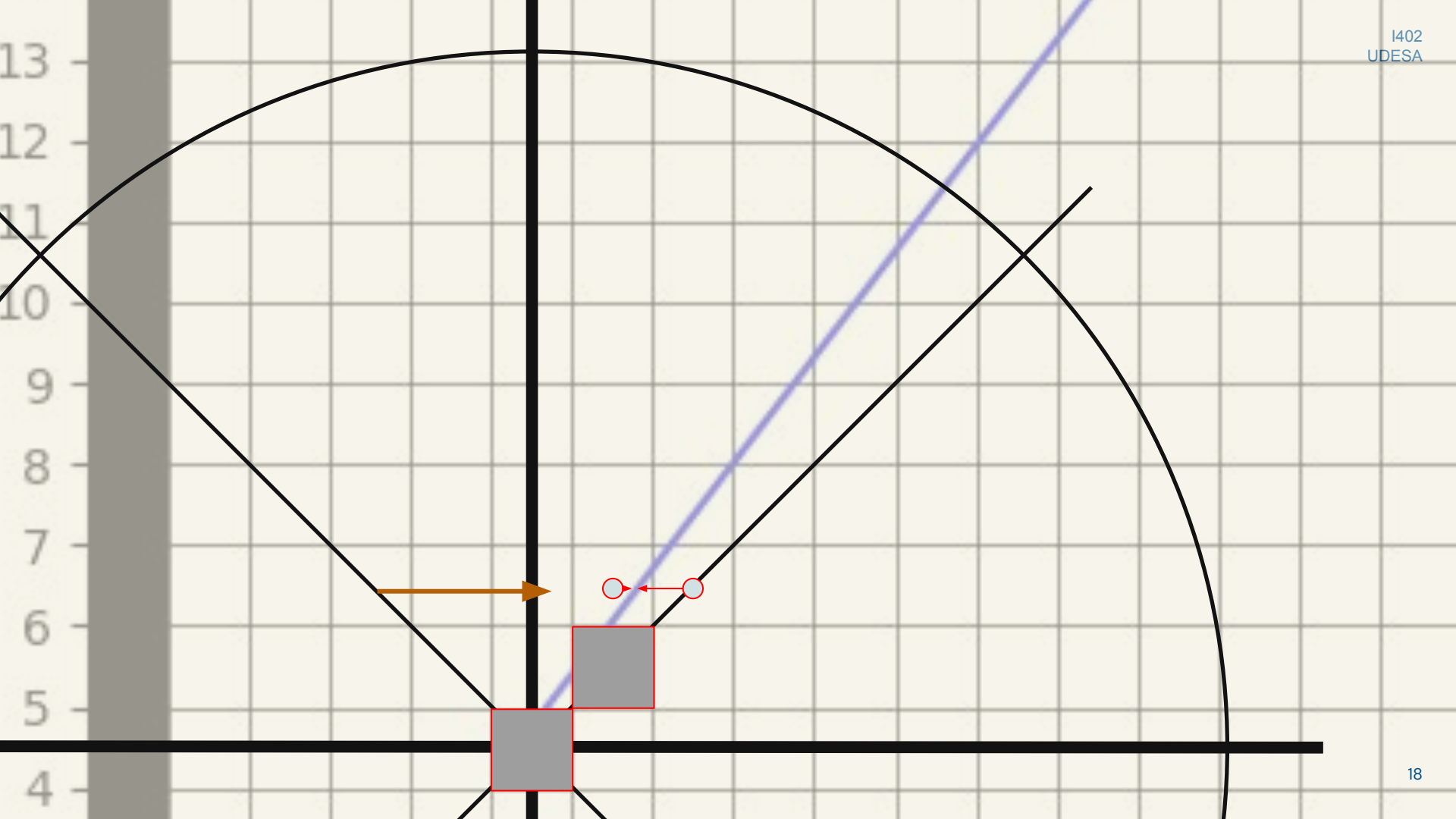
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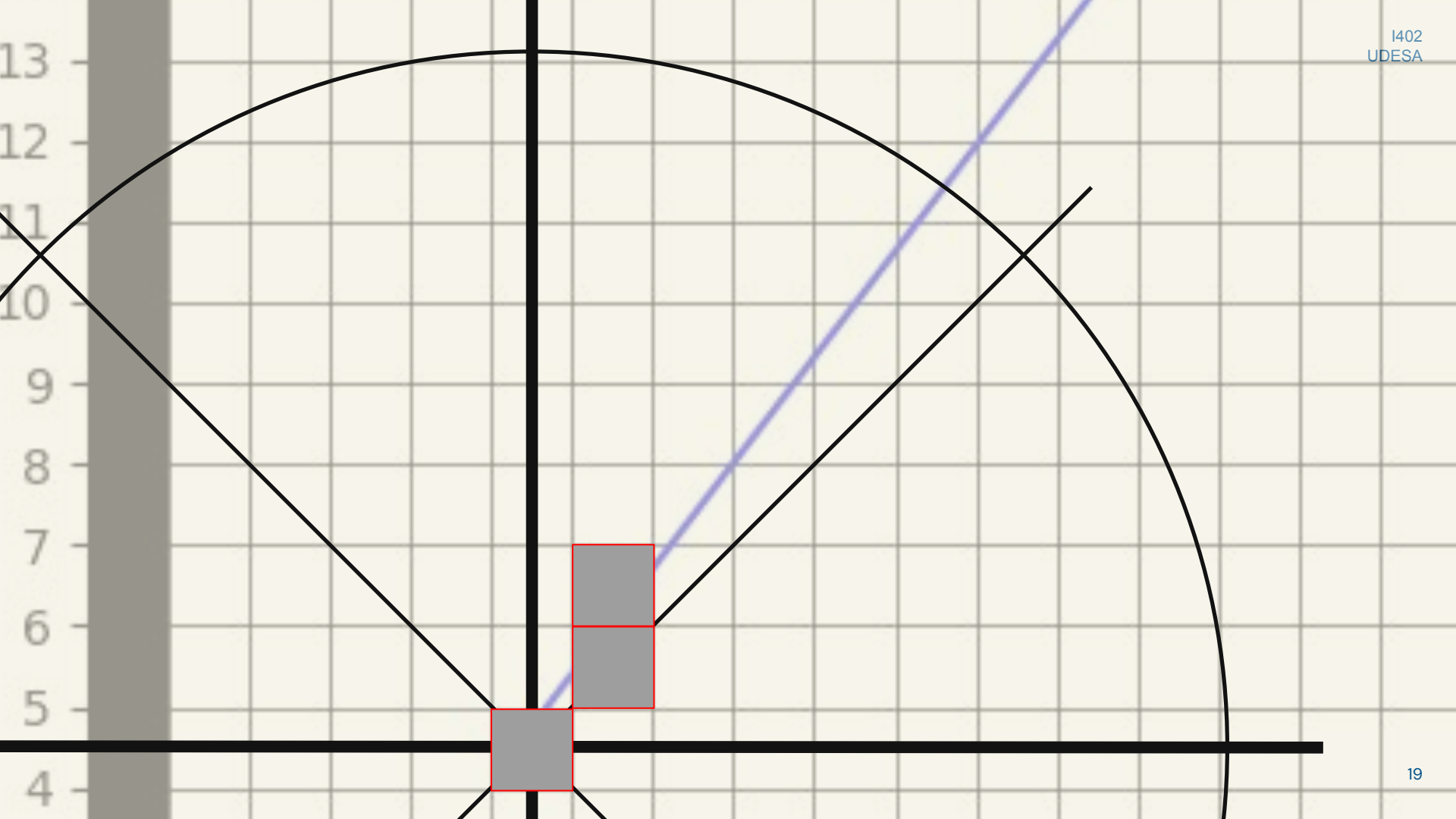
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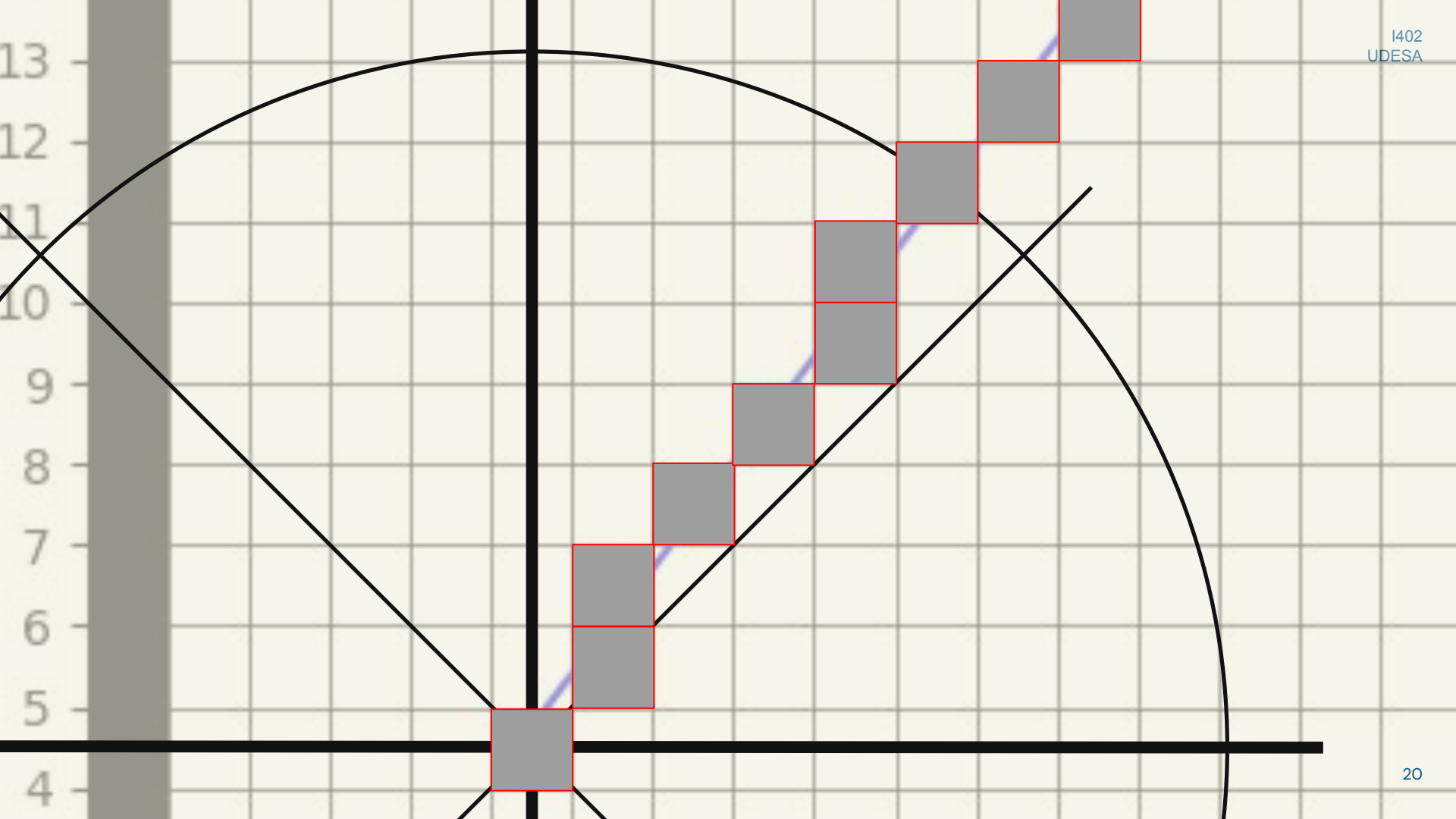
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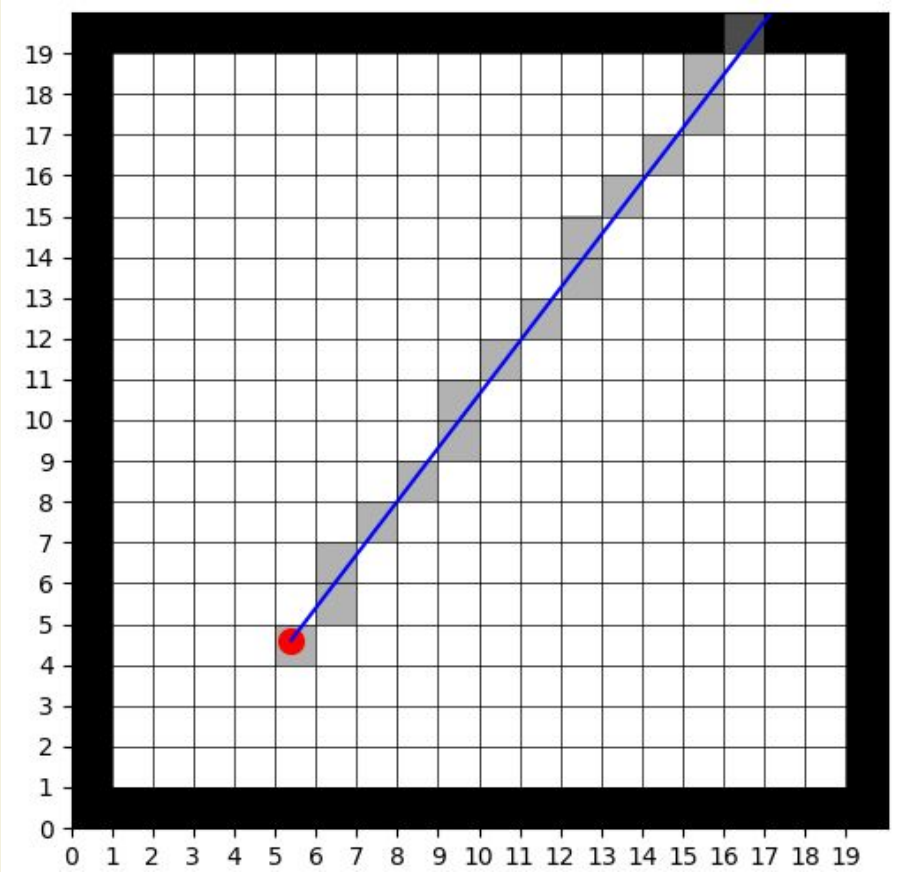
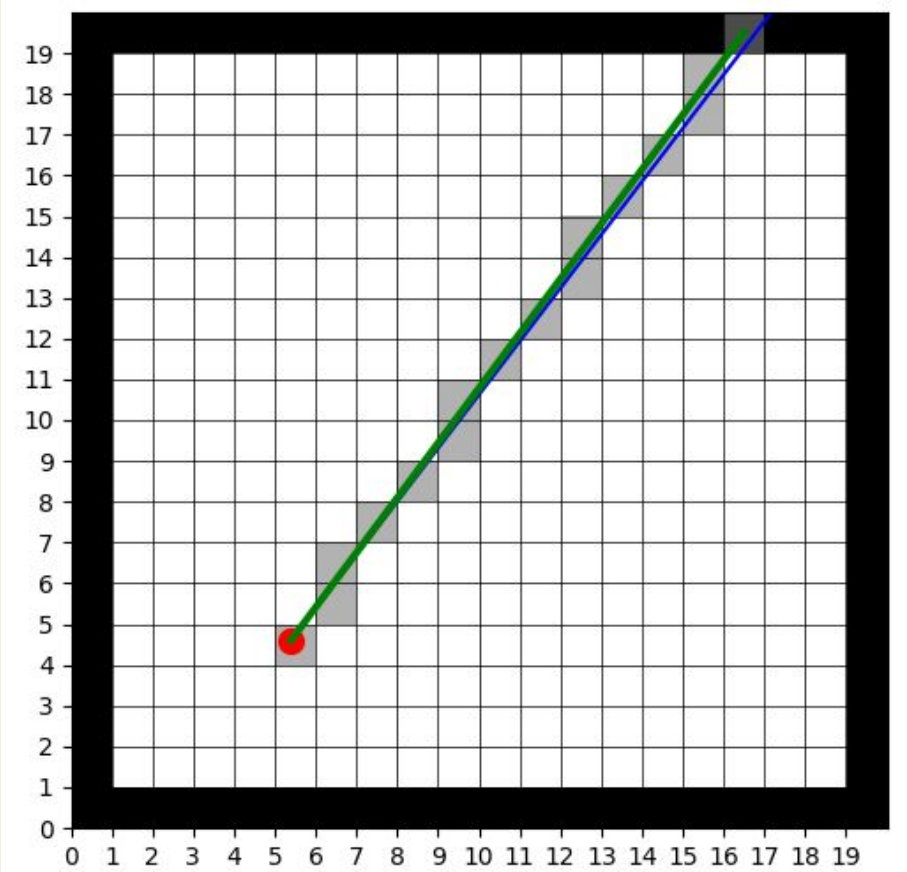
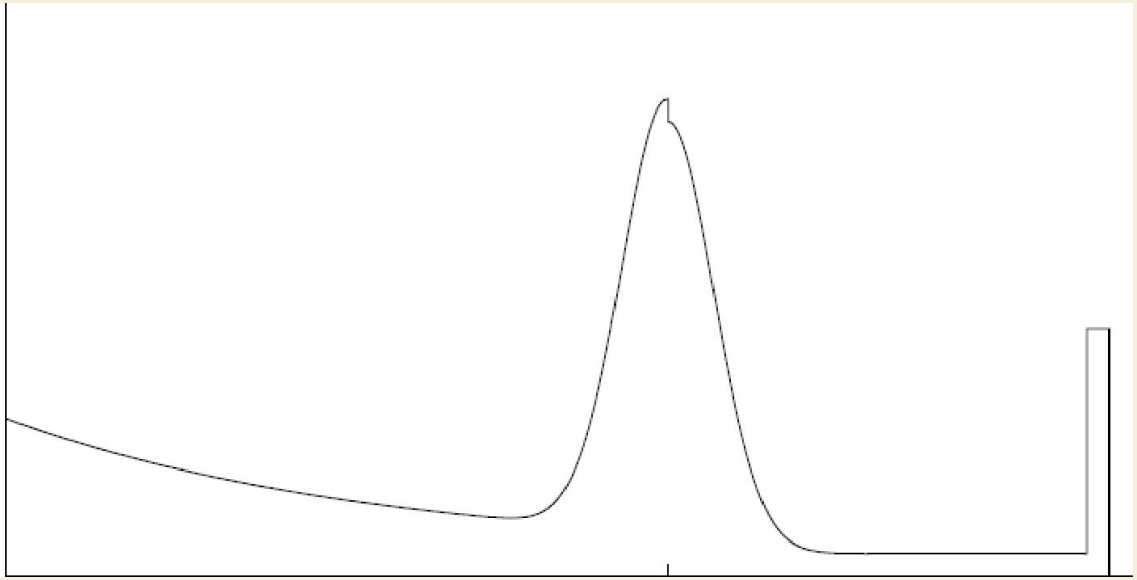
Bresenham Lines



Bresenham Lines



Modelos de Mediciones

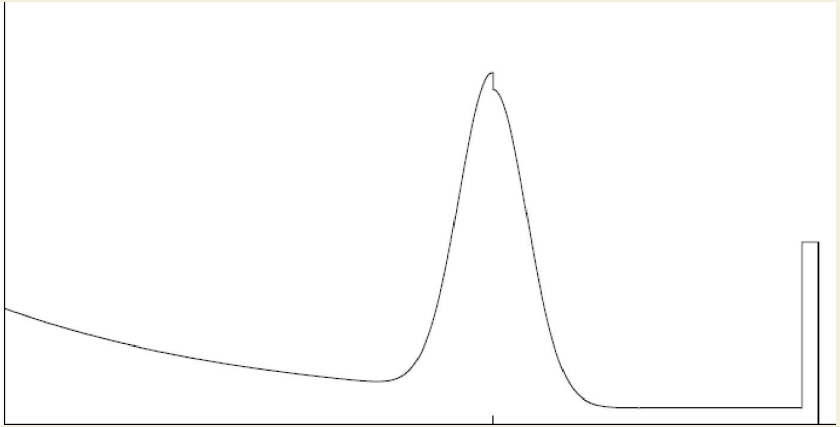


¿Cómo me afecta la resolución?

Modelos de Mediciones

$$P(z_t \mid x_t)$$

Ojo



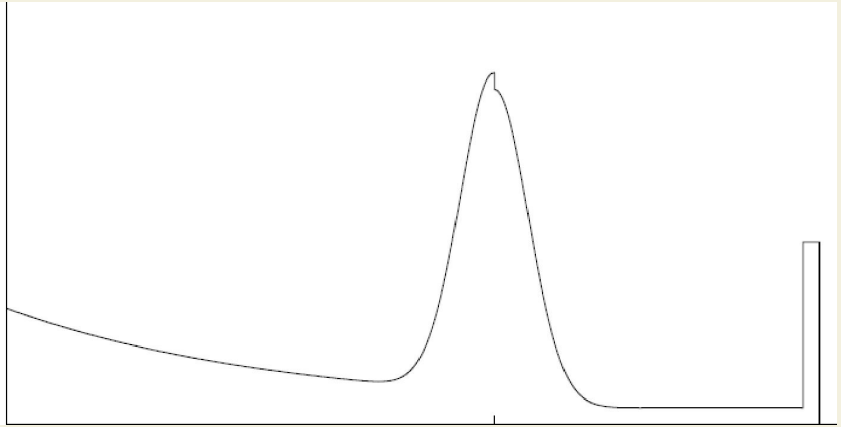
$$P(z \mid x, m) = \begin{pmatrix} \alpha_{\text{hit}} \\ \alpha_{\text{unexp}} \\ \alpha_{\text{max}} \\ \alpha_{\text{rand}} \end{pmatrix}^T \cdot \begin{pmatrix} P_{\text{hit}}(z \mid x, m) \\ P_{\text{unexp}}(z \mid x, m) \\ P_{\text{max}}(z \mid x, m) \\ P_{\text{rand}}(z \mid x, m) \end{pmatrix}$$

¿Que pasa si hay mucho ruido random? ¿Y mucho movimiento en el entorno? ¿Y si no confío mucho en mi mapa?

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Modelos de Mediciones

Qué problemas tiene este modelo? Hint: son 2



Algorithm beam_range_finder_model(z_t, x_t, m):

$q = 1$

for $k = 1$ *to* K *do*

compute z_t^{k*} *for the measurement* z_t^k *using ray casting*

$p = z_{\text{hit}} \cdot p_{\text{hit}}(z_t^k \mid x_t, m) + z_{\text{short}} \cdot p_{\text{short}}(z_t^k \mid x_t, m)$

$\quad + z_{\max} \cdot p_{\max}(z_t^k \mid x_t, m) + z_{\text{rand}} \cdot p_{\text{rand}}(z_t^k \mid x_t, m)$

$q = q \cdot p$

return q

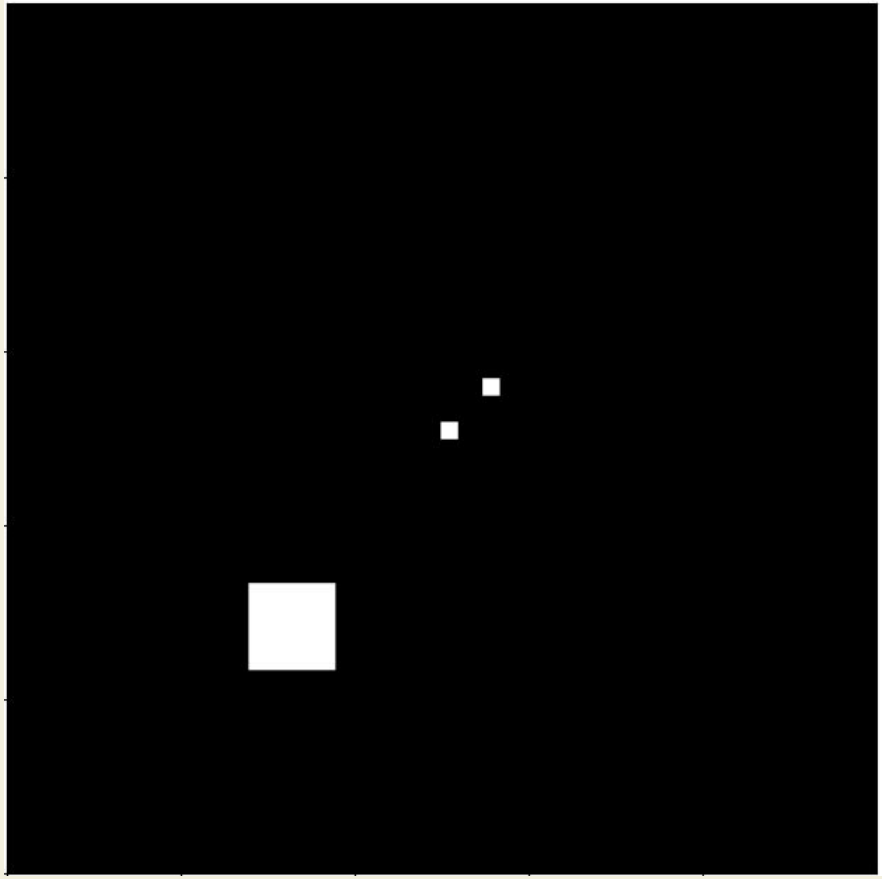
Modelos de Mediciones Likelihood Field

Occupancy grid map

Likelihood field

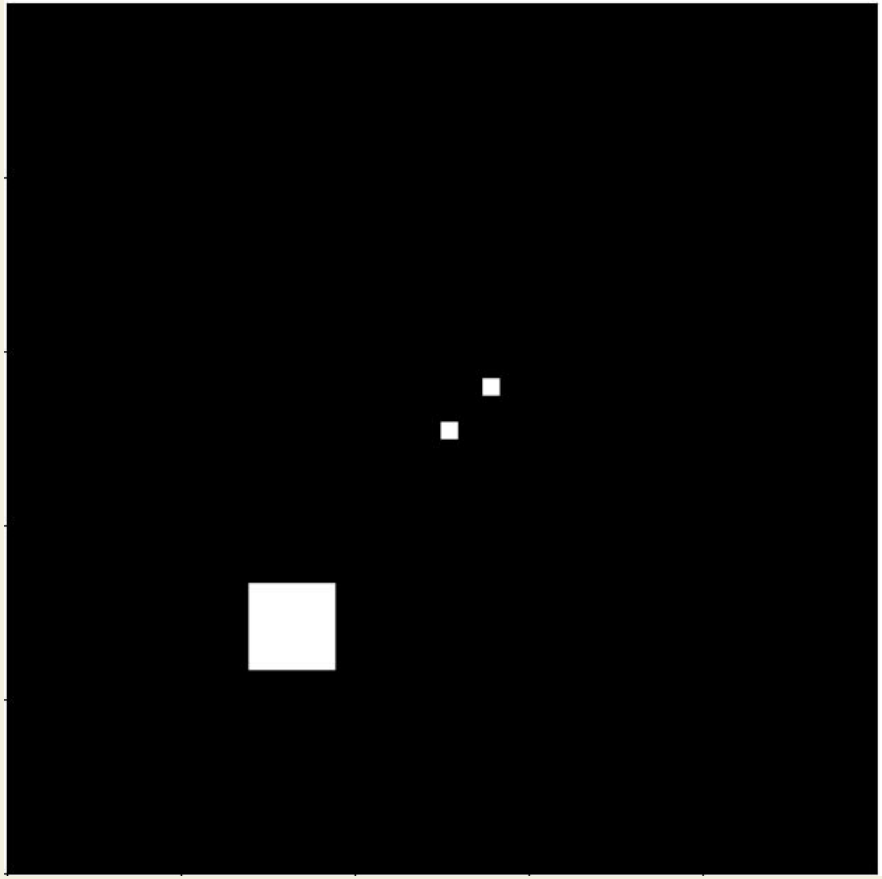
Modelos de Mediciones Likelihood Field

No tiene porqué ser un mapa 100% certero. Puede ser un mapa que “creo” que es así



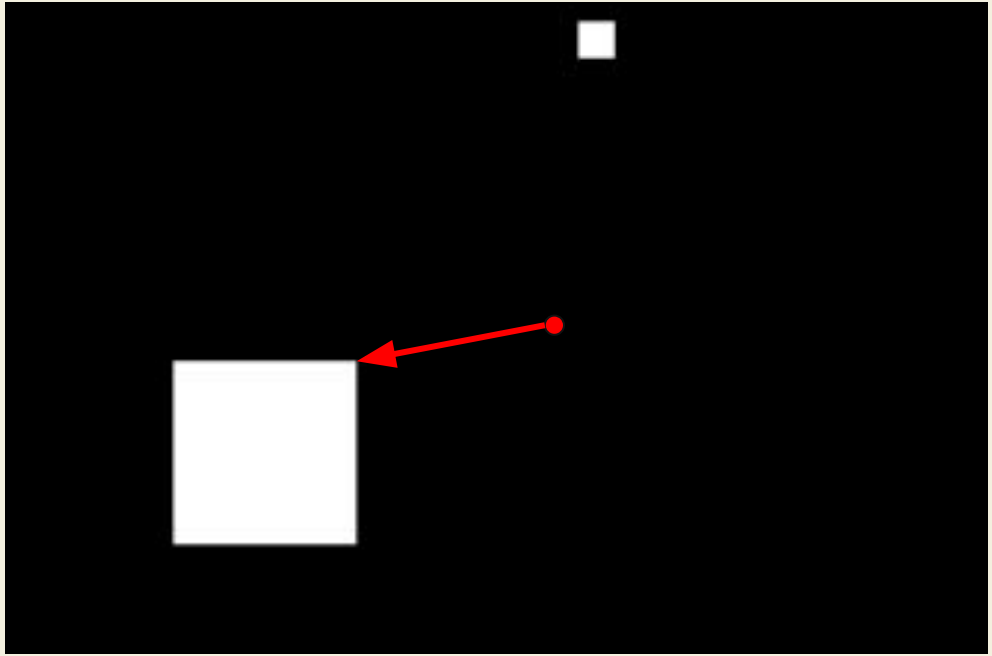
Modelos de Mediciones Likelihood Field

Para cada punto del espacio vacío puedo calcular su “likelihood” usando la distancia más corta hasta un punto ocupado y calculando una gaussiana con esa distancia

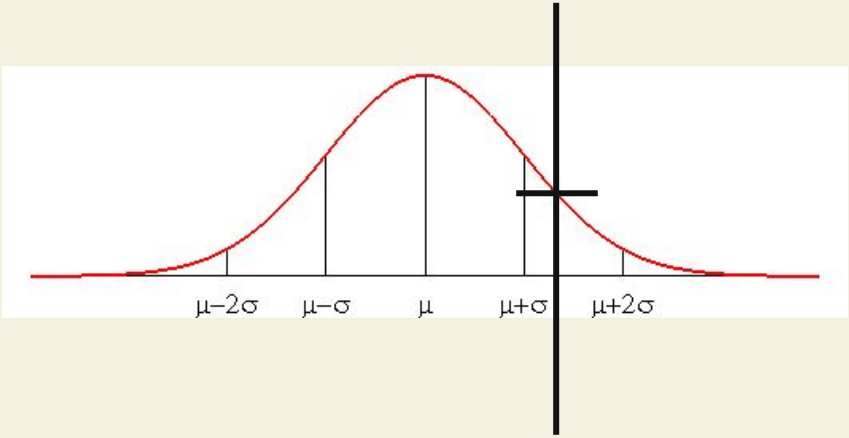


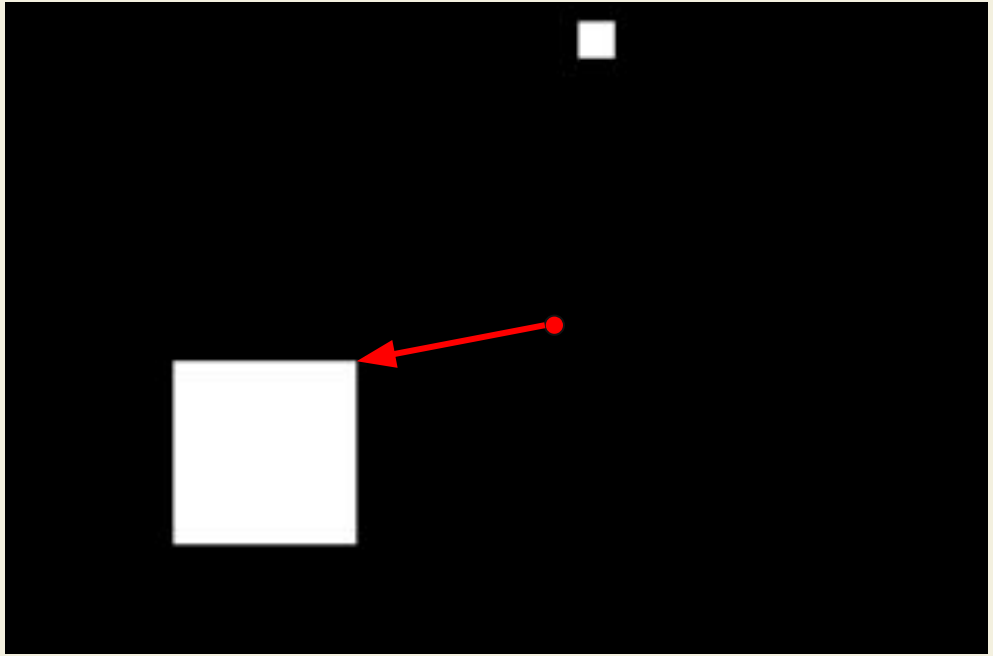
Modelos de Mediciones Likelihood Field

Para cada punto del espacio vacío puedo calcular su “likelihood” usando la distancia más corta hasta un punto ocupado y calculando una gaussiana con esa distancia

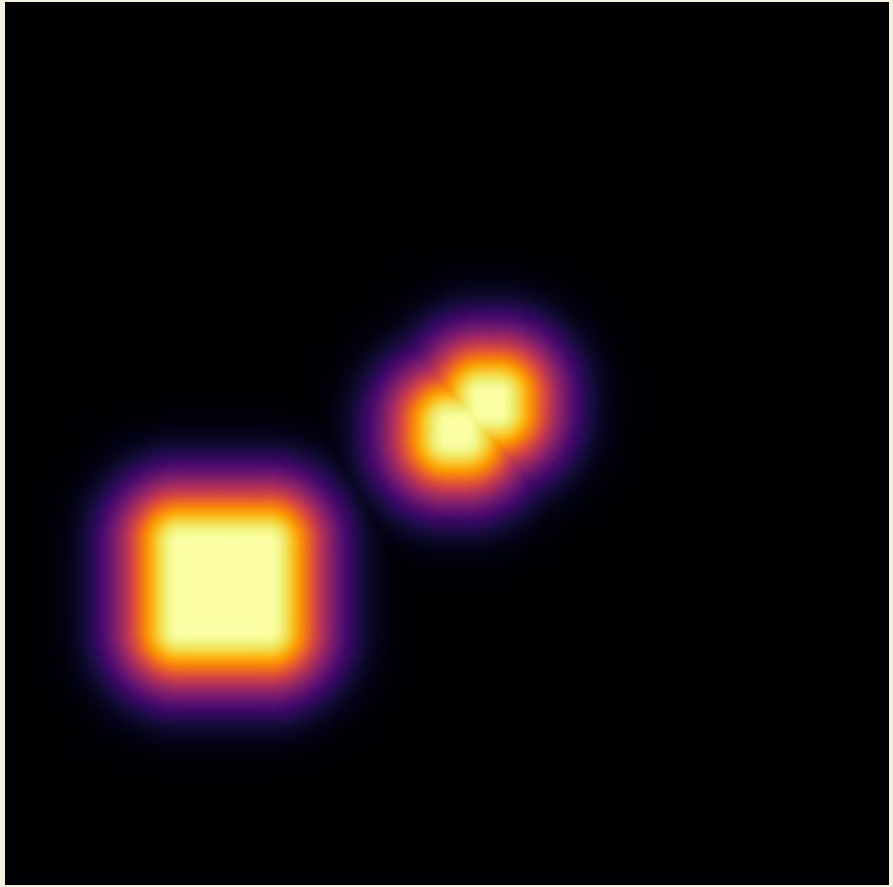


Modelos de Mediciones Likelihood Field





Modelos de Mediciones
Likelihood Field



Modelos de Mediciones Likelihood Field

Algorithm likelihood_field_range_finder_model(z_t, x_t, m)**:**

$q = 1$

for all k *do*

  if $z_t^k \neq z_{\max}$

    $x_{z_t^k} = x + x_{k,\text{sens}} \cos\theta - y_{k,\text{sens}} \sin\theta + z_t^k \cos(\theta + \theta_{k,\text{sens}})$

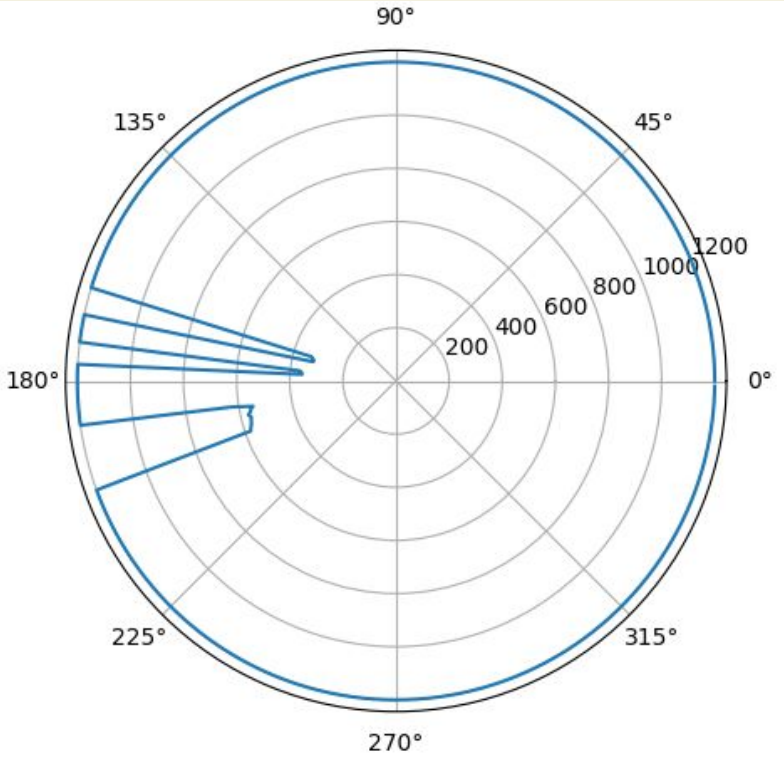
    $y_{z_t^k} = y + y_{k,\text{sens}} \cos\theta + x_{k,\text{sens}} \sin\theta + z_t^k \sin(\theta + \theta_{k,\text{sens}})$

    $dist = \min_{x',y'} \left\{ \sqrt{(x_{z_t^k} - x')^2 + (y_{z_t^k} - y')^2} \,\middle|\, \langle x', y' \rangle \text{ occupied in } m \right\}$

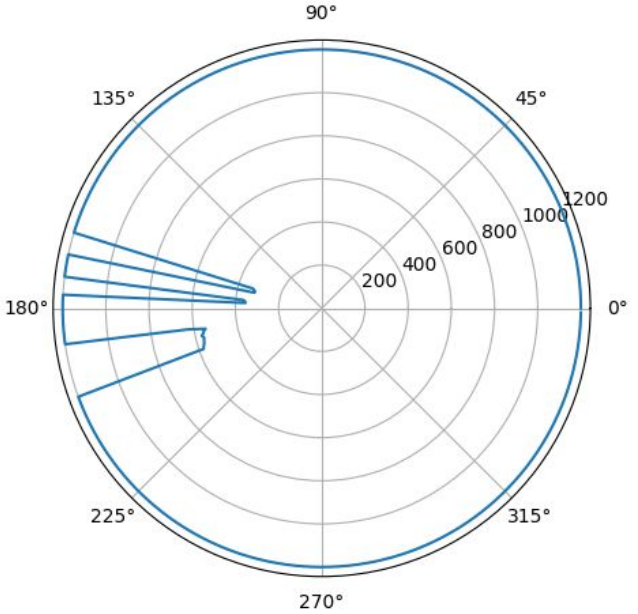
    $q = q \cdot \left(z_{\text{hit}} \cdot \mathbf{prob}(dist, \sigma_{\text{hit}}) + \frac{z_{\text{random}}}{z_{\max}} \right)$

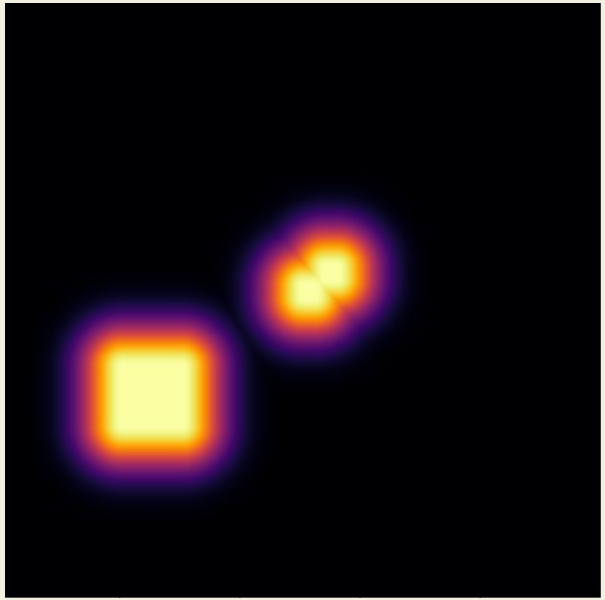
return q

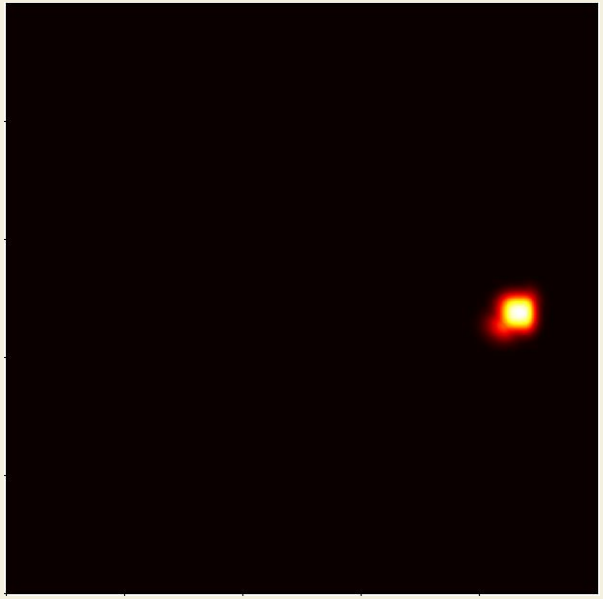
Modelos de Mediciones
Likelihood Field



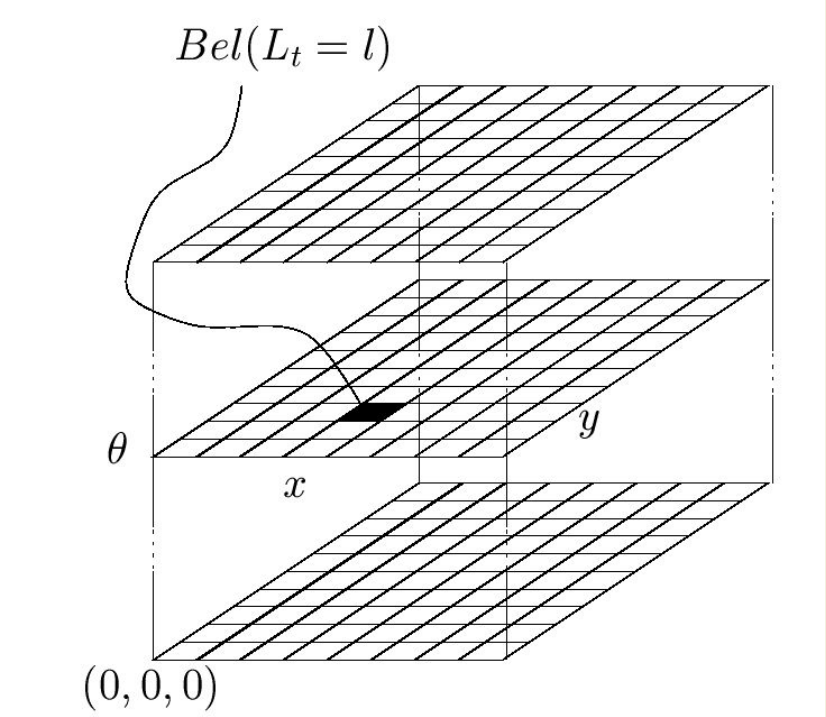
Modelos de Mediciones Likelihood Field







Discretización del Espacio



Algorithm Discrete_Bayes_filter$(\{p_{k,t-1}\}, u_t, z_t)$**:**

for all k *do*

$$\bar{p}_{k,t} = \sum_i p(X_t = x_k \mid u_t, X_{t-1} = x_i)\, p_{i,t-1}$$

$$p_{k,t} = \eta\, p(z_t \mid X_t = x_k)\, \bar{p}_{k,t}$$

endfor

return $\{p_{k,t}\}$

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Discretización del Espacio

Al ejecutar “avanzar un paso” el resultado será el siguiente:

- el 25 % de las veces, el robot no se moverá
- el 50 % de las veces, el robot avanzará una celda
- el 25 % de las veces, el robot se moverá dos celdas